

The Largest Common Ind Set

Problem: Find $I \in \mathcal{I}_1 \cap \mathcal{I}_2$ of max size.

The matroids $M_1 = (E, \mathcal{I}_1)$, $M_2 = (E, \mathcal{I}_2)$

are specified via an oracle to test membership in each $\mathcal{I}_1, \mathcal{I}_2$ respectively.

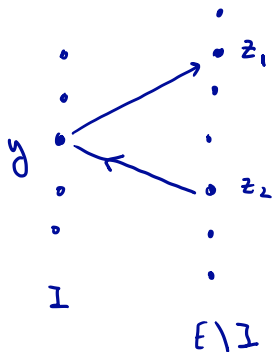
($\mathcal{I}_i, i=1,2$ need not be enumerated).

We want to find $\max: |I|$
efficiently in $|E|$

A Directed Graph

For $M_i = (E, \mathcal{I}_i)$, $i=1,2$

For $I \in \mathcal{I}_1 \cap \mathcal{I}_2$, define a
directed graph: $\textcircled{1} (I) = (E, A(I))$
 M_1, M_2



$(y, z_1) \in A(I)$.ff
 $I - y + z_1 \in \mathcal{I}_1$

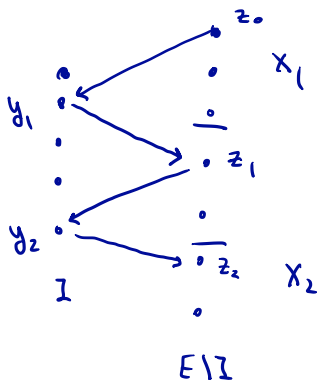
$(z_2, y) \in A(I)$.ff
 $I - y + z_2 \in \mathcal{I}_2$

Augmenting Algorithm - Case I

Given $\mathcal{X}_1, \mathcal{X}_2$ and $I \in \mathcal{X}_1 \cap \mathcal{X}_2$

$$\mathcal{X}_1 \triangleq \{z \in E \setminus I : I + z \in \mathcal{X}_1\}$$

$$\mathcal{X}_2 \triangleq \{z \in E \setminus I : I + z \in \mathcal{X}_2\}$$



Case I : $D_{M_1, M_2}(I)$

has a $\mathcal{X}_1 - \mathcal{X}_2$.

Let P be a shortest such path.

Claim: $I' = I \Delta VP$,

$I' \in \mathcal{X}_1 \cap \mathcal{X}_2$.

Claim says: $I - y_1 - y_2 + z_0 + z_1 + z_2 \in \underline{\mathcal{X}_1 \cap \mathcal{X}_2}$

Case I Augments

Thm: $I' = I \Delta VP \in \mathcal{I}_1 \cap \mathcal{I}_2$.

Note: b/c P is shortest $x_1 - x_2$ path, the arcs in P leaving I form the only matching in $D_{M_1}(I)$ on the nodes $VP - z_0$. $\textcircled{**}$

Also, $\forall z \in VP \setminus I - z_0$, $I + z \notin \mathcal{I}_1$, else there would be a shortest path. $\textcircled{**}$

Let $J = I \Delta VP - z_0$. In picture: $J = I - y_1 - y_2 + z_1 + z_2$.

$|I| = |J|$. Also: $r_{M_1}(I \cup J) = r_{M_1}(I) = |I|$, this follows from $\textcircled{*}$ since $z_1, z_2 \in \text{Span}(I)$.

Also from $\textcircled{**}$ $\exists$ unique matching in $D_{M_1}(J)$ on $I \Delta J$.

By 2nd Corollary in Lecture on Matroids, Part VII, $I + z_0 \in \mathcal{I}$,
 $\Rightarrow J + z_0 \in \mathcal{I}_1$.



Augmenting Algorithm - Case II



Proof — continued —